

B.TECH FINAL YEAR - MAJOR PROJECT PRESENTATION

RoomRadar

AI-Powered Room Rental and Booking Platform

A full-stack MERN web application for room discovery, landlord listings, secure bookings, real-time chat, admin moderation and AI assistance.

B.Tech Final Year Major Project Presentation

Presented by: [Your Name] Guided by: [Guide Name]

Department of [Branch] | [College Name] | 2026

Presentation Overview

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Introduction and Problem Statement

Domain

RoomRadar belongs to the prop-tech domain and solves room discovery, listing management, booking and communication problems through one digital platform.

Background

Students, travellers and working professionals often depend on scattered offline brokers, incomplete listings and slow communication while searching for rental rooms.

Problem

The existing process lacks verified room information, fast filtering, transparent booking status, secure document handling and one trusted communication channel.

Objective

To build a role-based rental platform where travellers search and book rooms, landlords manage listings, and admins moderate trust and safety workflows.

Literature and Technology Review

Online Rental Marketplaces

Digital marketplaces improve reach and convenience, but local rental systems still need stronger verification, filtering and dispute support.

Geospatial Search

Location-based search, radius filtering and sorting improve relevance near colleges, offices and preferred areas.

Real-Time Messaging

Socket-based chat improves user confidence by enabling direct inquiry handling and booking-related notifications.

Trust and Safety

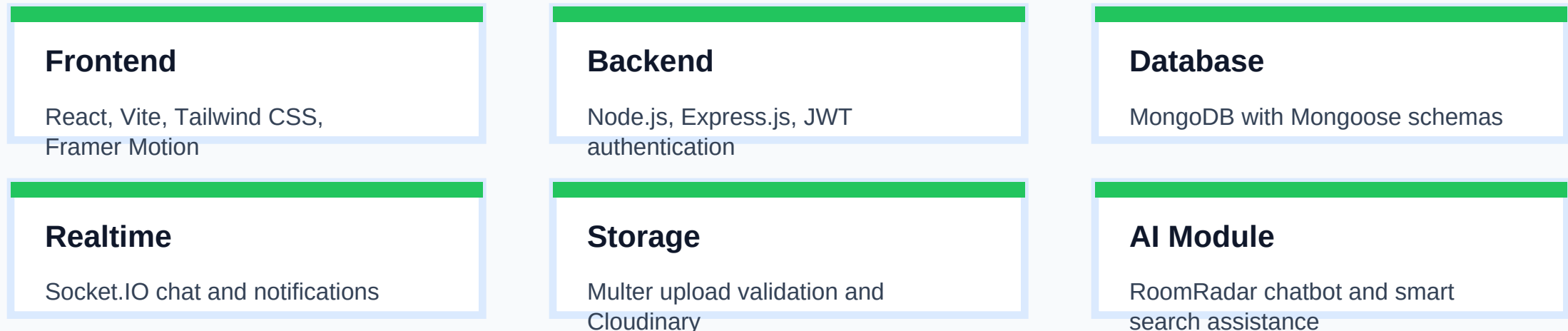
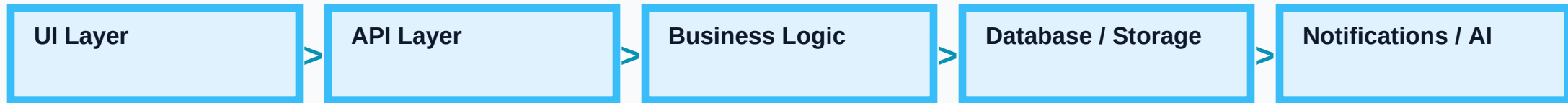
Role-based access, verified listings, reviews, upload validation and admin moderation reduce fraud, spam and disputes.

Proposed System / Methodology



The methodology follows an iterative full-stack development approach: define modules, implement UI and APIs, connect workflows, harden edge cases and verify production readiness.

System Architecture



Implementation and Results

Working modules include traveller search, room details, booking requests, landlord listing management, admin moderation, inbox, chatbot and support tickets.

3**Role-Based Portals****End-to-End** Search to Booking Flow**Realtime****Chat and Notifications****Verified****Build and Syntax Checks**

Latest verification: client production build passed and server syntax checks completed without errors.

Testing and Evaluation

Test Case / Module	Expected / Actual Output	Status
TC-01: Login and Role Routing	Correct dashboard opens	PASS
TC-02: Location Search and Filters	Rooms update by city, radius and filters	PASS
TC-03: Booking Request	Valid room, ID proof, terms and future date required	PASS
TC-04: Chat and Inquiry	Inquiry and conversation flow connected	PASS
TC-05: Upload Validation	Unsafe files rejected by type and size checks	PASS

Conclusion and Future Work

Conclusion

RoomRadar provides a complete digital rental workflow for room discovery, listing management, booking, chat and support. The platform improves transparency through verified listings, reviews, protected booking steps and admin moderation.

Future Work

Deploy on cloud infrastructure, integrate online payment gateway, develop Android/iOS apps, add AI recommendations, multilingual chatbot support and analytics dashboards.

References

- [1] React Documentation and Vite Documentation: <https://react.dev/> | <https://vite.dev/>
- [2] Express.js Documentation: <https://expressjs.com/>
- [3] MongoDB and Mongoose Documentation: <https://www.mongodb.com/docs/> | <https://mongoosejs.com/docs/>
- [4] Socket.IO Documentation: <https://socket.io/docs/>
- [5] Cloudinary Documentation: <https://cloudinary.com/documentation>
- [6] OWASP Web Application Security Guidelines: <https://owasp.org/>

Thank You!

Questions and Discussions Welcome

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